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Investigating dementia patterns by machine learning and graph-based marker analysis for multiclass dementia diagnosis

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Abstract

Dementia is a debilitating disease that leads to a gradual loss of memory and other cognitive abilities and tends to develop in people over 60 years of age. The number of people worldwide with dementia is expected to increase from 57.4 million in 2019 to 131.5 million in 2050. Diagnosis of dementia is a critical task, and some biomarkers and psychological and demographic measures have been used to diagnose dementia clinically. The advent of artificial intelligence (AI) has created new opportunities to improve diagnostic accuracy. In this study, we propose an approach based on machine learning and graph analysis to identify important patterns and diagnostic markers for dementia. The novel contributions of this study include a descriptive probability density function (PDF) analysis to examine the distribution of clinically relevant markers across diagnostic groups, providing exploratory insights into group-wise feature behavior. A comprehensive evaluation of ten machine learning (ML) models is conducted under strict subject-level data separation to identify robust classification performance. In addition, feature ranking is assessed using three complementary strategies (Random Forest importance, Chi-Square statistics, and Boruta ranking) to evaluate the consistency and robustness of marker relevance. Out of ten ML models evaluated, the Random Forest achieved the highest accuracy of 94.37%, with a macro F1-score of 86.96%. The result of the ML model is further validated using a graphical analysis of critical markers. The results of this study might help to improve the diagnostic accuracy.

Keywords Dementia, Clinical markers, Graph analysis, Machine learning, Decision rules, Knowledge discovery

1 Introduction

Dementia is a clinical syndrome resulting from neurodegeneration [1, 2]. Dementia can be described as an irreversibly progressive decline in cognitive ability and independent living capacity [1]. The most prevalent underlying pathologies for dementia are Alzheimer's Disease (AD), vascular dementia, Lewy body dementia, and frontotemporal dementia [2]. Memory loss and impaired judgment are traits of dementia, leading to gradual



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deterioration of an individual's ability to carry out even the most basic daily tasks [3]. Dementia symptoms tend to worsen with age; however, they can appear at any stage in life. Aging is a leading risk factor for prevalent neurodegenerative diseases [4]. According to Mantovani et al. and Levine et al., there will be approximately 74.7 million dementia cases worldwide by 2030, rising to 131.5 million by 2050 [1, 3].

Diagnosis of dementia is a crucial task since this disease's progression can be slow, and in the majority of cases, the deterioration occurs gradually and does not result in diagnosis until advanced dementia is evident [5]. Some clinical biomarkers (i.e., Normalized Whole Brain Volume (nWBV), Clinical Dementia Rating (CDR), Atlas Scaling Factor (ASF), Estimated Total Intracranial Volume (eTIV)) and some psychological and demographic markers (i.e., Education Level (EDUC), Socio-Economic Status (SES), Mini-Mental State Examination (MMSE)) have been used over the years in the diagnosis of this disease [6].

The advent of AI has revolutionized the field of medicine and healthcare, providing numerous benefits to clinicians and medical professionals [7–9]. AI technologies, especially machine learning (ML) algorithms, and deep neural networks have enabled researchers to analyze enormous amounts of medical data and identify complex patterns that were difficult to discern using conventional diagnostic methods [7, 10]. AI can increase the speed and accuracy of medical assessments, reduce the likelihood of misdiagnosis, and improve the quality of treatment. Integration of AI in medicine is a promising trend [11]. In recent years, graph-based approaches have shown that structural and functional connectivity patterns can enhance classification accuracy and interpretability [12]. The main objective of this study is to present an improved model based on ML and Graph analysis that helps to detect dementia at an early stage.

Previous studies [6, 13–25] have employed advanced techniques to diagnose dementia. Machine learning and deep learning (DL) models have been trained on tabular and image datasets to classify different types of dementia. In addition, some studies have utilized graph-based approaches for classification. However, these existing studies have some key gaps, such as a need for proper feature analysis and a decision support system. Furthermore, clinicians require insights and guidelines to aid in their diagnosis.

This proposed study aims to address the limitations of previous studies and assist clinicians by uncovering the impact of different features on individual class labels. This approach distinguishes our work from previous studies using the OASIS-2 dataset, where the primary focus has been on improving model performance. Our study, in contrast, integrates exploratory data analysis to provide insights into the relationships between key clinical markers and dementia stages. The main contribution of this study lies in the combination of tabular machine learning, feature selection techniques, and graphical visualization, which together provide a comprehensive and interpretable approach for exploring dementia diagnosis. Additionally, we employ Probability Density Function (PDF) analysis and correlation visualization to enhance the interpretability of the results, providing insights into how each marker contributes to the classification of dementia stages. Moreover, the Neo4j tool is employed to visualize graphs, illustrating their respective impact on class labels. The integration of graph-based analysis allows us to visualize and better understand the relationships between clinical markers. These innovative contributions are detailed below:

- The probability density function (PDF) graph is analysed to identify latent patterns in diagnostic markers for dementia.
- Exploratory analysis of the ML models is performed, validating the importance of integrating clinicians' findings and AI.
- Feature ranking analysis is provided to identify key markers for dementia classification.
- Critical diagnostic markers are presented and analysed graphically.

2 Related works

The advancement of AI has been extensively observed in automated dementia diagnosis. Researchers developed several techniques and models for the analysis of dementia markers and disorders. Both ML and DL models are gaining much attention for effective diagnosis. This section aims to review the state-of-the-art methods for dementia prediction, identify these techniques, and finally compare their outcome with those of this proposed study.

In the first study, Basheer et al. [6] proposed a deep neural network approach to classify dementia using the OASIS dataset. They modified a capsule network to build a framework and attained an accuracy of 92.39%. Kavitha et al. used several ML techniques, including Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting, and Voting classifiers, utilizing feature ranking and training with different models, resulting in a test accuracy of 83% for the OASIS dataset [13]. Alzheimer's disease is the most common reason for dementia. El-Sappagh et al. [14] analyzed AD progression, applying information fusion with heterogeneous ensemble classifiers and using the ADNI dataset. Their proposed Stacking with the XGB ensemble model obtained an accuracy of 89.15%. This study emphasized model tuning for accuracy improvement, while our study also investigated the correlation between the data. Additionally, recent studies in computational biology have also demonstrated that hybrid feature representations combined with ensemble learning methods, such as XGBoost-Sumo [26], can significantly enhance predictive performance. Furthermore, explainable AI methods, such as SHAP, have been widely adopted to enhance interpretability in high-dimensional biological prediction tasks [27].

In another study, Murugan et al. [21] proposed DEMNET, a deep learning method for early detection of Alzheimer's disease and dementia from MRI images using a Kaggle dataset. They used the SMOTE technique to balance the dataset and provided individual Alzheimer's disease probability maps. DEMNET achieved 95.23% accuracy on the Kaggle dataset and 84.83% on the ADNI dataset. Mohammed et al. [22] used DL and hybrid methods, finding that the hybrid AlexNet+SVM model achieved the best result of 94.8% on the OASIS dataset. Lim et al. [24] developed a smart healthcare system with a deep neural network, achieving 99% accuracy on a private dataset using scaled principal component analysis to reduce data dimensions, though they did not analyze marker identification or data correlation. Hu et al. [23] proposed a DL-based approach to classify frontotemporal dementia and Alzheimer's using the ADNI and NIFD datasets, achieving 91.83% accuracy and utilizing a gradient visualization method for differential diagnosis. Similar trends toward deep neural architectures have also been observed in other biomedical domains, such as the PSSM-Sumo framework [28], which integrates evolutionary feature encoding with DNN-based classification. Additionally, hybrid deep learning

architectures have also been applied to sequential data for combining transformer-based embeddings with temporal convolutional networks to improve classification performance [29].

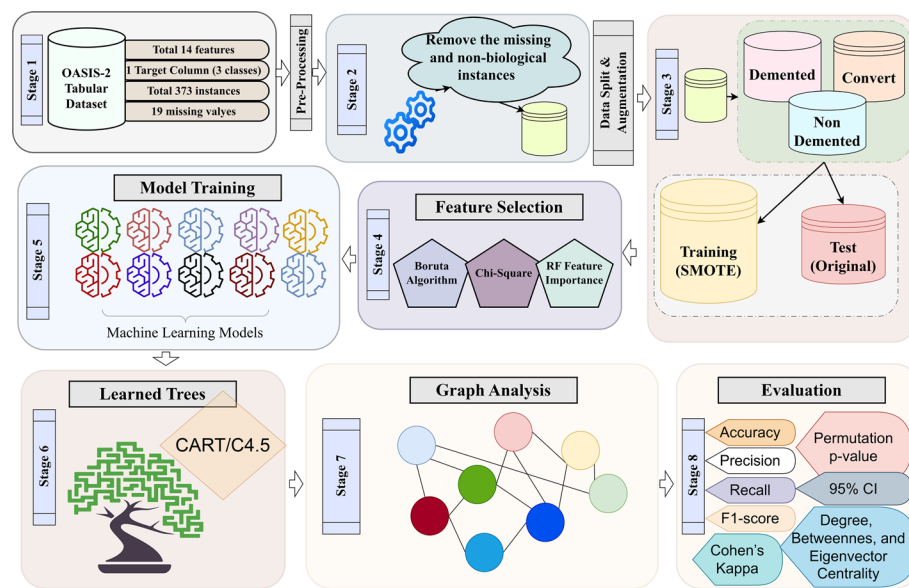
In another study, Moguilner et al. [15] investigated the reliability of a multimodal computational framework for characterizing neurodegenerative diseases in underrepresented populations, capturing complex associations between nonlinear brain connectivity features and pathophysiological and cognitive profiles for AD through the application of multiple ML approaches. Khan et al. [16] proposed a framework for classifying dementia on the OASIS dataset. The framework is a multimodal-based ML approach with different feature ranking methods. The highest classification accuracy of 87% was achieved with the random forest ML model. Hashmi et al. [25] proposed a system based on reinforcement learning and neural networks for early diagnosis of dementia utilizing the OASIS dataset. They utilized deep reinforcement learning to represent a person's dementia disease risk accurately, which also addressed the class imbalance issue. Their proposed approach increases the trial accuracy by 6%, precision by 9%, recall by 13%, and F-score by 9–10%, respectively.

The recent graph-based application and analysis significantly contribute to finding the relevant patterns and markers of different diseases [17]. GCN approaches have recently shown their efficacy in diagnosing diseases associated with dementia [18]. Shang et al. [19] developed a novel graph capsule convolutional network (GCCN) architecture to predict the transition from mild cognitive impairment (MCI) to dementia. Their research collected the target risk genes and considered the brain ROI as nodes to construct the graph. The graph capsules captured the discriminative information from the gene factors and provided vital insight into the progression of MCI to dementia. Alzheimer's disease is the most common cause of dementia, and Song et al. [18] proposed a graph neural network technique for diagnosing the disease. They constructed the graph by randomly selecting the samples, introduced a metric-based meta-learning method to reduce the complexity and enhance the learning capability from the samples, and finally performed the node classification tasks for unknown labels. Hett et al. [20] proposed a graph-based grading approach for Alzheimer's disease prediction. They analyzed the brain structures from the image data, constructed the graph based on the subject's similarity, and used a cascade classifier to predict the disease. These studies suggest the significant contributions of graphs for determining dementia utilizing essential markers.

After conducting a thorough review of the extant literature, several significant limitations have been identified that must be addressed to enhance the diagnostic system's transparency and reliability. Most studies place a strong emphasis on model development for predictive tasks, failing to conduct proper feature analysis to explore the correlation and impact of the various features. Additionally, the absence of graphical visualization for critical markers and a decision support system is noteworthy, as these components are crucial for clinicians to gain insights from these methodologies. Our study aims to address these limitations systematically. Table 1 offers a comparative analysis of these studies.

Table 1 Comparative analysis of existing studies on dementia prediction

Study	Dataset	Feature ranking	PDF analysis	AI models	Marker-based knowledge discovery
[6]	OASIS	X	X	✓	X
[13]	OASIS	✓	✓	✓	X
[14]	ADNI	✓	X	✓	X
[15]	ADNI + NIFD	X	X	✓	X
[16]	OASIS	✓	✓	✓	X
[17]	ADNI	X	X	✓	X
[18]	TADPOLE	X	X	✓	X
[19]	ADNI	X	X	✓	X
[20]	ADNI	✓	X	✓	X
[21]	Kaggle + ADNI	X	X	✓	X
[22]	OASIS + MRI datasets	✓	X	✓	X
[23]	ADNI + NIFD	X	X	✓	X
[24]	Custom Dataset	X	X	✓	X
[25]	OASIS + ADNI	✓	X	✓	X
Ours	OASIS-2	✓	✓	✓	✓

**Fig. 1** Proposed pipeline for dementia prediction: integration of broad feature analysis and machine learning approach with the investigation of graphs

3 Methods

This section presents the proposed methodology for dementia prediction and analysis. We designed a multi-stage framework that integrates data preprocessing, feature selection, machine learning-based classification, interpretable decision tree analysis, and graph-based modeling. The methodology is structured to evaluate multiple learning models, analyze feature contributions, and examine relationships among markers. The following subsections describe each stage of the framework in detail. Figure 1 represents the methodology of this study.

3.1 Dataset description

In our research, we utilized the Open Access Series of Imaging Studies (OASIS-2) [30] dataset, the second version of the OASIS dataset. This dataset includes information from 150 people (aged 60 to 96), with more than two scanning sessions. Each session was at least a year apart, and the average time between visits was 719 days. The dataset contains information such as: Subject ID, MRI ID, Group, Visit, MR delay, Sex, Age, Social Economic Status (SES), Education level (EDUC), Mini-Mental State Examination (MMSE), Clinical Dementia Rating (CDR), estimated Total Intracranial Volume (e-TIV), normalized Whole Brain Volume (n-WBV) and Atlas Scaling Factor (ASF). Every patient has a unique subject ID and MRI ID (for each scanning session). The CDR feature denotes the Clinical Dementia Rating for each patient at each visit. The range of the CDR is from 0 to 3. In the dataset we have utilized, the value of CDR can be 0, 0.5, or 1, depending on the class. The ASF column denotes the Atlas Scaling Factor, which is a factor that converts brain and space inside the skull to an atlas target. The nWBV feature indicates the size of a person's brain in relation to the size of their body, and the eTIV column is the measurement to estimate the brain's total volume.

3.2 Dataset preprocessing

The OASIS-2 tabular dataset contains 373 instances in three classes: Demented (146), Non-demented (190), and Converted (37). Among the 373 instances, 19 instances contain missing values. Additionally, there are 17 missing values in the SES column and 2 in the MMSE column. After removing these instances from the dataset, 354 instances remain, with 127, 190, and 37 samples in the respective classes. As there is a substantial disparity between the majority class (non-demented) and the minority class (converted), the machine learning models will be biased toward the majority class when data are provided. To address this issue, we applied the Synthetic Minority Oversampling Method (SMOTE). However, to preserve the integrity of the fairness of the evaluation, the augmentation was only applied to the Training set after the dataset split. Several variables available in the OASIS dataset, including MR Delay, ASF, and Visit number, represent acquisition timing, atlas scaling, or longitudinal indexing rather than biological mechanisms of dementia. These variables were therefore excluded from all diagnostic interpretation, feature importance analysis, and model reasoning. Estimated total intracranial volume (eTIV) was treated strictly as a normalization covariate and was not used as a predictive or decision-driving feature. All biologically meaningful inference in this study is restricted to MMSE, normalized whole brain volume (nWBV), Age, EDUC, and SES. The preprocessing step is visualized in Fig. 2. Table 2 provides an overview of the dataset distribution.

3.3 Analysis of critical markers

Graphs of the probability density function (PDF) are indispensable for data analysis and interpretation, particularly in medicine [31]. These visualizations provide valuable insight into data distribution and facilitate the identification of patterns, trends, and hidden information. In the context of dementia diagnosis, PDF graphs can assist in identifying markers that strongly contribute to the disease's development and progression. In our study, we analyze the PDF graphs to identify significant characteristics, such as peaks, modes, and tails, that can provide crucial information about the disease's

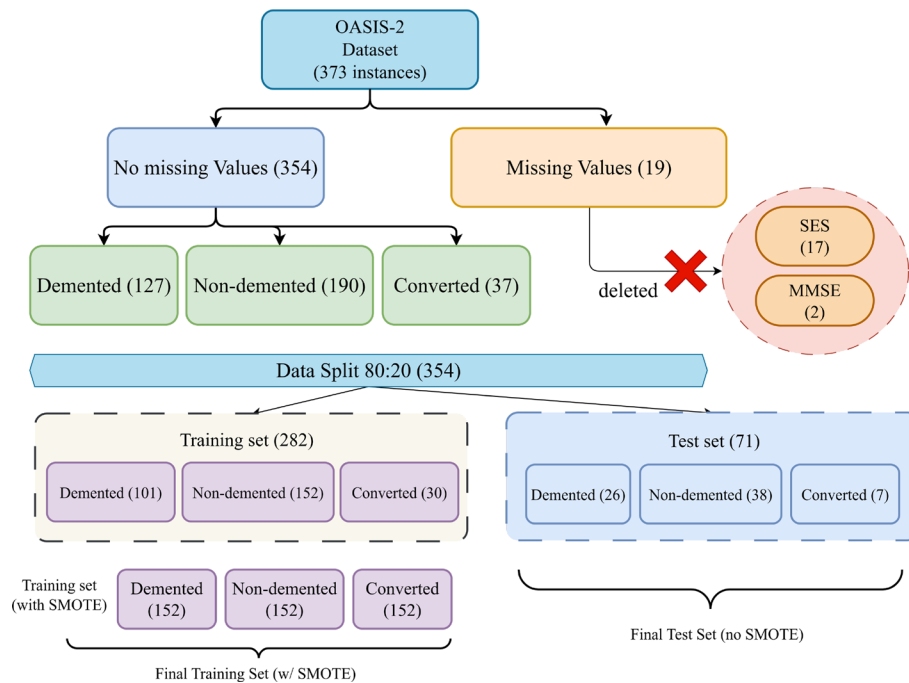


Fig. 2 Dataset breakdown for model training and testing. The missing values in the SES and MMSE were discarded. Only the final training set was augmented with SMOTE; the test set was untouched

Table 2 Complete dataset and preprocessing summary

Stage	Demented	Non-demented	Converted	Total
Original samples per class	146	190	37	373
After removing missing samples	127	190	37	354
Training set (before SMOTE)	101	152	30	283
Test set (real)	26	38	7	71
Training set after SMOTE	152	152	152	456

After removing the missing samples, the dataset was split into an 80:20 ratio for training and testing, respectively. Later, SMOTE was used only on the training set to increase the sample size in the training set

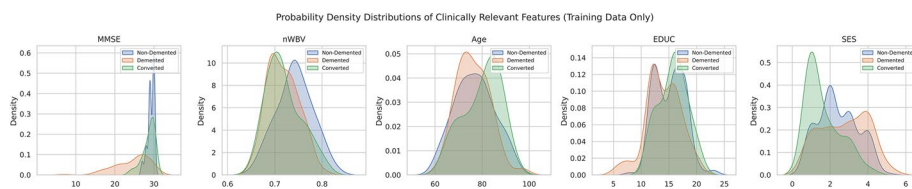


Fig. 3 Probability density graph of MMSE, nWBV, Age, EDUC, and SES

underlying pathology. Utilizing the PDF graph ensures that the data are correct and reliable, making it an essential tool for ensuring data quality. We constructed probability density graphs to display the distribution of the significant markers that indicate a patient's current condition. These bell-shaped curve graphs provide insight into the frequency of the parameters in a specific range relating to the different classes. Figure 3 depicts the PDF graph of five important clinical markers in our study that are MMSE, nWBV, Age, EDUC, and SES.

Figure 3 illustrates the subjects' mini-mental state (MMSE) condition, which is usually assessed using a set of questions. The plot reflects that a lower value of the MMSE

suggests the patient has dementia. As the value increases, the risk of patients being affected by dementia declines. The literature agrees with this, reporting that the lower values are found for severely demented patients while the patients with mild or no dementia have a higher MMS [32]. Additionally, nWBV refers to the brain volume of the subjects. The demented patient has the highest density but less volume than non-demented or converted patients. A demented person's brain eventually starts to shrink [30].

The distribution of age demonstrates a general shift toward higher ages among demented and converted groups. Nevertheless, the overlapping density curves indicate that age alone does not provide a clear diagnostic boundary and must be contextualized within a broader clinical framework. Also, the subjects with less education tend to be affected more by dementia. For a lower education level, there is a higher probability of being demented. Outcomes of the previous studies also support that lower education is a potential risk factor for dementia [33].

SES is an indicator that refers to various socioeconomic factors, including income and status. From Fig. 3, we can see that a progressive increase in the SES value increases the chance of being affected by dementia. From existing literature, it was found that a lower socioeconomic status increases the potential for the development of dementia [34]. The graph illustrates that dementia development increases slowly with increasing SES values.

PDF graphs give us valuable insights into the markers and how they can distinguish each class's values. These markers assist our approach to finding the exact path to the three classes of our study.

4 Machine learning analysis

4.1 Machine learning model

This section describes our ML-adopted approach to validate the proposed methodology. In this study, we focused on ten machine learning methods. Below is a brief discussion of these models:

Decision Tree: The Decision Tree algorithm is a hierarchical supervised learning model introduced for its reliable performance. A decision tree operates sequentially and can be used to solve classification and regression problems. Various techniques, including ID3, C4.5, C5.0, and CART, can be used to learn the decision from the trees.

Random Forest: RF is an effective algorithm, known for its remarkable performance in classification and regression tasks. It employs several random and uncorrelated decision trees to develop a forest. It is considered an efficient technique since it demands less computational cost compared to the classical boosting or bagging approaches. The RF classifier is more efficient in selecting the training samples and noise reduction in training datasets.

Gradient Boosting: Gradient Boosting (GB) is a Boosting algorithm that builds the weak learner from a small training sample and achieves the best results in the shortest time. This method addresses the loss function optimization and reduces the overfitting tendency.

$$D = \{(x_i, y_i)\} (|D| = n, x_i \in R^m, y_i \in R) \quad (1)$$

$$L(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (2)$$

Here, D is the given dataset and n is the number of samples, and m denotes the number of features. Equation (2) is employed to minimize regularization. Here, $L(\phi)$ denotes the minimized loss function, l is the residual difference between the real and predicted value. Ω is a regularization term that penalizes model complexity and (f_k) refers to the k th index weak learner.

XGBoost: XGBoost is an improved version of the gradient boosting algorithm that accelerates the speed and performance of the model. This method can efficiently optimize the execution process. The XGBoost model also offers excellent accuracy in multi-class classification tasks.

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2 \quad (3)$$

In Eq. (3), γ and λ are the penalty coefficients, T is the number of leaves, and ω is the sum of leaf weight.

AdaBoost: Adaptive Boosting (AdaBoost) is a boosting-based ensemble method that iteratively trains weak classifiers by assigning higher weights to misclassified samples in subsequent iterations. This adaptive reweighting mechanism allows the model to focus on difficult instances and improves overall predictive performance.

Support Vector Machine (RBF Kernel): Support Vector Machine (SVM) with a radial basis function (RBF) kernel is a nonlinear classifier that constructs an optimal decision boundary by maximizing the margin between classes in a transformed feature space. The RBF kernel enables effective modeling of nonlinear class boundaries, making SVM suitable for structured clinical datasets.

Logistic Regression (Multinomial): Multinomial Logistic Regression (MLR) is a probabilistic linear classifier that models the relationship between multiple categorical outcomes and input features using the softmax function. Despite its simplicity, it provides strong interpretability and serves as a reliable baseline for multiclass classification tasks.

k-Nearest Neighbors: The k-Nearest Neighbors (kNN) algorithm is a distance-based, non-parametric classifier that assigns class labels based on the majority vote among the k closest samples in the feature space. Although sensitive to noise and feature scaling, kNN offers intuitive decision boundaries and serves as a useful comparative baseline.

Naïve Bayes (Gaussian): Gaussian Naïve Bayes (NB) is a probabilistic classifier based on Bayes' theorem and the assumption of conditional independence among features. Despite its simplicity, it remains effective for small datasets and provides computationally efficient probabilistic predictions.

Artificial Neural Network (MLP): The Multilayer Perceptron (MLP) is a feedforward artificial neural network composed of multiple fully connected layers that learn hierarchical feature representations through nonlinear transformations. The MLP captures complex and nonlinear relationships among input features by adjusting network weights via backpropagation. The forward propagation of an MLP can be expressed as shown in Equation (4):

$$h^{(l)} = f \left(W^{(l)} h^{(l-1)} + b^{(l)} \right) \quad (4)$$

where $h^{(l)}$ denotes the activation of the l -th hidden layer, $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector, respectively, and $f(\cdot)$ is a nonlinear activation function.

4.2 Feature ranking analysis

This study aims to discover significant markers to predict dementia. It was mentioned earlier that the dataset consists of various features, including clinical, psychological, and demographic data. Three different feature selection techniques, (i) Boruta Algorithm, (ii) Chi-Square, and (iii) Random Forest Feature Importance, are applied to find the optimal feature set.

- (i) *The Boruta algorithm* is based on a similar idea to the Random Forest classifier. It shuffles the attributes randomly, runs the classifier, and computes the Z scores. The procedure is repeated until all features are either accepted or rejected. The Boruta algorithm compares real features against randomized shadow features within a Random Forest framework, thereby identifying variables that consistently contribute information beyond noise. However, Boruta may retain correlated features and does not differentiate causal relevance from predictive utility.
- (ii) *Random forest feature importance* calculates the mean decreased impurity of the attributes to select the essential features. The higher the purity value for attribute tree leaves, the higher the importance, and this is recursively done for all the attributes. Random Forest importance reflects each feature's contribution to impurity reduction within an ensemble-based, multivariate classification framework. While this approach captures non-linear interactions between variables, it is known to exhibit bias toward continuous features and correlated predictors. On the other hand, RF importance is interpreted as a measure of predictive contribution.
- (iii) *The Chi-Square test* provides a model-agnostic assessment of marginal association between individual features and the diagnostic group. Although computationally efficient and statistically interpretable, this method does not account for feature interactions and may underestimate the relevance of variables that exert influence primarily in multivariate contexts. Chi-square is a statistical tool used to compare actual and observed values. It is of great use in finding correlations between the categorical variables.

$$\text{Chi-square} = \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} \quad (5)$$

Table 3 summarizes feature relevance assessed using three complementary feature-ranking strategies. Random Forest (RF) importance, Chi-Square statistics, and Boruta ranking are applied to the original dataset without synthetic oversampling. These methods report the feature relevance and assess the robustness of clinically meaningful markers across different selection criteria. Importantly, feature rankings are not interpreted as biological or clinical causation, but rather as indicators of predictive and statistical relevance within the studied dataset.

Table 3 Feature ranking results obtained using Random Forest importance, Chi-Square statistics, and Boruta ranking

Feature	RF importance	Chi-square	Boruta rank
MMSE	0.374	3.116	1
nWBV	0.255	3.150	1
Age	0.177	0.559	1
EDUC	0.105	1.050	2
SES	0.090	5.628	1

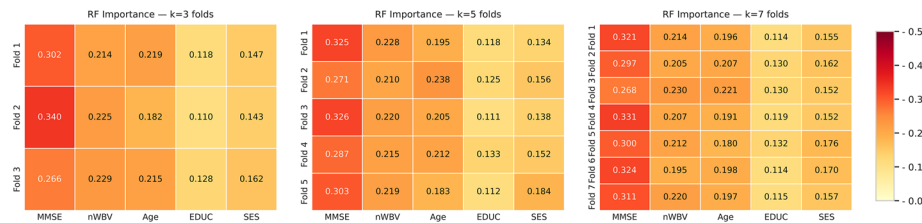


Fig. 4 Feature importance stability across stratified k-fold cross-validation

Table 4 Test Accuracy and Macro F1-score of evaluated models across three splits: 60:40, 70:30, and 80:20 for train and test set

Model	Accuracy			Macro F1			
	60:40	70:30	80:20	60:40	70:30	80:20	Average
Random forest	0.8803	0.9089	0.9437	0.7698	0.7994	0.8696	0.8130
ANN-MLP	0.8820	0.9019	0.9085	0.7269	0.7639	0.7952	0.7620
XGBoost	0.8486	0.8949	0.9049	0.7078	0.7678	0.7904	0.7553
Gradient boosting	0.8468	0.8972	0.8873	0.6944	0.7755	0.7713	0.7471
KNN	0.7623	0.8364	0.8768	0.6729	0.7350	0.7779	0.7286
Decision tree	0.8187	0.8598	0.8768	0.6579	0.6956	0.7486	0.7007
SVM-RBF	0.8398	0.8481	0.8662	0.7252	0.7119	0.7300	0.7224
Logistic regression	0.8151	0.8738	0.8592	0.7164	0.7540	0.7441	0.7382
Naive bayes	0.8715	0.8411	0.8415	0.7531	0.7093	0.6964	0.7196
AdaBoost	0.7236	0.6752	0.6303	0.6148	0.5655	0.5681	0.5828

The results are averaged over SMOTE k

To examine the robustness of the selected features, we conducted a stability analysis using Random Forest feature importance across stratified k-fold cross-validation ($k = 3, 5, \text{ and } 7$). Figure 4 illustrates the heatmap of feature importance values across different folds. In this procedure, the training set was repeatedly partitioned into stratified folds, and a Random Forest model was trained on each fold after applying SMOTE within the training partition to maintain class balance. Feature importance values were recorded for each fold, and their distribution was compared across folds to assess consistency. The results indicate that the ranking of the most influential markers, particularly MMSE and normalized whole brain volume (nWBV), remains consistent across folds. This stability suggests that the selected features provide robust predictive signals within the dataset.

5 Results analysis

All of our experiments were conducted with a fixed random seed (42) for data splitting, model initialization, and SMOTE oversampling. The dataset was partitioned using a class-wise fixed protocol for the 60:40, 70:30, and 80:20 train-test splits. We applied SMOTE only to the training set after splitting, with k values of 3, 5, 7, and 9. For all split configurations, we ensured subject-level separation so that data from the same individual did not appear in both training and test sets. All models were trained using only the five feature sets: MMSE, nWBV, Age, EDUC, and SES. The experiments were implemented in Python 3.9 using scikit-learn 1.2.2 and imbalanced-learn 0.10.1.

Table 4 summarizes the test accuracy and macro F1-score of all 10 models across three train-test splits, averaged over SMOTE k values. Overall, tree-based methods (RF) and neural networks consistently performed the best. RF achieved the highest average macro F1 (0.810), closely followed by ANN-MLP. XGBoost also showed solid performance,

with test accuracies above 87% and balanced precision-recall. Simpler models, such as AdaBoost and KNN, lagged especially on smaller splits. In general, all models improved as the training split increased, with the 80–20 split yielding the highest test scores.

Most top-performing models achieved their best results on the 80–20 split, with ensemble methods (RF, GB, XGBoost) and neural networks showing peak performance under this configuration. Table 5 summarizes the best-performing configuration for each model, including the train-test split and SMOTE k -value that achieved the highest test accuracy. The 95% confidence intervals reveal that Random Forest not only achieves the highest accuracy (0.9437) but also maintains a relatively narrow interval [0.903 to –0.965]. Gradient Boosting and XGBoost exhibit similarly tight bounds. In contrast, simpler models such as AdaBoost and Naive Bayes reached their peak performance on different splits and SMOTE values, with AdaBoost showing notably wider confidence intervals [0.721 to –0.748] relative to its point estimate (0.7465), reflecting greater uncertainty in its predictions. While KNN's accuracy CI is moderately wide [0.898 to –0.907], its recall CI [0.803 to –0.836] is comparatively tighter. These bootstrap-derived intervals (1000 iterations) provide a comprehensive view of the statistical reliability of our experiments.

Table 5 Best split and SMOTE configuration per model with corresponding performance metrics

Model	Split	SMOTE _k	Classification performance				95% Confidence intervals			Brier score
			Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	
RF	80:20	3	0.9437	0.9039	0.8484	0.8696	0.903 to –0.965	0.882 to –0.905	0.823 to –0.851	0.056
GB	70:30	5	0.9159	0.8449	0.7734	0.7939	0.863 to –0.955	0.819 to –0.847	0.744 to –0.777	0.071
ANN-MLP	80:20	3	0.9155	0.8288	0.7879	0.8034	0.896–0.940	0.801–0.830	0.759 to –0.792	0.103
DT	80:20	3	0.9155	0.8287	0.7920	0.8054	0.898 to –0.918	0.802 to –0.828	0.764 to –0.796	0.155
XGBoost	80:20	9	0.9155	0.8288	0.7879	0.8034	0.893 to –0.925	0.801 to –0.830	0.759 to –0.792	0.079
KNN	80:20	3	0.9155	0.8268	0.8308	0.8286	0.885 to –0.925	0.799 to –0.829	0.803 to –0.836	0.161
MLR	70:30	7	0.8879	0.7597	0.7285	0.7380	0.828 to –0.940	0.731 to –0.763	0.698 to –0.732	0.118
SVM-RBF	80:20	3	0.8873	0.7535	0.7356	0.7412	0.865 to –0.891	0.725 to –0.757	0.706 to –0.739	0.131
NB	60:40	5	0.8803	0.7737	0.7542	0.7626	0.857 to –0.882	0.745 to –0.777	0.724 to –0.758	0.143
AdaBoost	60:40	9	0.7465	0.6484	0.6438	0.6349	0.721 to –0.748	0.619 to –0.651	0.613 to –0.647	0.198

95% confidence intervals were estimated using non-parametric bootstrap resampling with 1000 iterations to quantify the variability of accuracy, precision, and recall (sensitivity) estimates. The macro multiclass Brier score is also reported for each model to assess probability calibration; lower values indicate better-calibrated outputs

Additionally, for probability calibration, we computed the multiclass Brier score for each model on its best-performing held-out test set using the predicted class probabilities. For classifiers that do not natively output calibrated probabilities (DT and KNN), we applied Platt scaling post-hoc before computing the score. Among the classifiers, RF achieves the lowest Brier score (0.056), which further confirms that its probability estimates are well-calibrated (in addition to being discriminative). Gradient Boosting (0.071) and XGBoost (0.079) also showed well-calibrated outputs, while AdaBoost (0.198) and KNN (0.161) showed comparatively weaker calibration.

5.1 Statistical model comparison

To assess the relative performance of the evaluated models (on the 80:20 train-test split), we computed key statistical metrics, Cohen's Kappa, and permutation p -values. These metrics quantify both overall agreement with the true labels and the likelihood of achieving the observed accuracy by chance. Table 6 summarizes the results for all ten models, including 95% confidence intervals for Kappa estimated via bootstrap resampling.

Among the models, Random Forest achieved the highest Cohen's Kappa (0.922) with a 95% confidence interval of [0.876, 0.958] and a highly significant permutation p -value (0.0005). Ensemble-based methods such as XGBoost and Gradient Boosting also performed strongly, with Kappa values around 0.898, 95% CIs of [0.842, 0.941], and highly significant p -values. The linear and probabilistic models, including Logistic Regression and Naive Bayes, achieved moderate agreement (Kappa 0.671 to 0.773) with confidence intervals that do not include zero ([0.543, 0.779] and [0.679, 0.851], respectively), and were significantly better than chance. In contrast, models such as ANN-MLP (Kappa 0.082, CI [0.000, 0.194]), KNN (Kappa 0.010, CI [0.000, 0.089]), and SVM-RBF (Kappa 0.076, CI [0.000, 0.187]) showed very low Kappa values with confidence intervals that include zero. This is consistent with their non-significant permutation p -values. AdaBoost achieved intermediate performance (Kappa 0.580, CI [0.472, 0.688]) with a significant p -value. For all models, we observe perfect alignment between the confidence intervals and p -values: every model with a p -value < 0.05 has a 95% confidence interval that excludes zero, and every model with a p -value ≥ 0.05 has a confidence interval that includes zero.

Table 6 Statistical comparison of model performance on the 80:20 split

Model	p -Value	Kappa	95% CI Kappa
Random forest	0.0005	0.922	[0.876, 0.958]
ANN-MLP	0.0980	0.082	[0.000, 0.194]
XGBoost	0.0005	0.898	[0.842, 0.941]
Gradient boosting	0.0005	0.898	[0.842, 0.941]
KNN	0.5000	0.010	[0.000, 0.089]
Decision tree	0.0005	0.776	[0.682, 0.854]
SVM-RBF	0.1684	0.076	[0.000, 0.187]
Logistic regression	0.0005	0.671	[0.543, 0.779]
Naive bayes	0.0005	0.773	[0.679, 0.851]
AdaBoost	0.0005	0.580	[0.472, 0.688]

All results are computed on the held-out test set. 95% Confidence Intervals for Cohen's Kappa were computed using non-parametric bootstrap random resampling with 1000 iterations

Table 7 Split-wise performance of the Random Forest classifier

Split	Accuracy	95% Accuracy CI	Macro Metrics			Permutation <i>p</i> -Value
			Prec	Recall	F1	
60:40	0.9195	0.872–0.960	0.8376	0.8225	0.8289	0.0005
70:30	0.9204	0.867–0.965	0.8609	0.7939	0.8139	0.0005
80:20	0.9437	0.903–0.965	0.9039	0.8484	0.8696	0.0005
Mean	0.9279	–	0.891	0.8216	0.8375	0.0005

Metrics are averaged over SMOTE configurations. 95% confidence intervals (CI) computed via bootstrap

Table 8 Per-class accuracy, precision and recall (sensitivity) for the Random Forest classifier on the 80:20 held-out test set (SMOTE $k = 3$, seed 42)

Class	Accuracy	Precision	Recall
Non-demented	0.965	0.960	0.974
Demented	0.944	0.930	0.924
Converted	0.922	0.822	0.647
Avg.	0.9437	0.9039	0.8484

The macro row reports the mean of per-class values

Table 9 k-Fold cross-validation (CV) results on the 80:20 split

k	CV Acc	CV Cohen's Kappa	<i>p</i> -Value
3	0.888	0.802	0.0005
5	0.892	0.807	0.0005
7	0.943	0.922	0.0005

Mean metrics are reported across folds. The (permutation) *p*-values computed on the 20% holdout

5.2 Performance analysis of the best model

Table 7 summarizes the performance of the Random Forest classifier across three train-test splits (60:40, 70:30, and 80:20). We computed all metrics exclusively on held-out test samples. Test accuracies were 91.95%, 92.04%, and 94.37% for the 60:40, 70:30, and 80:20 splits, respectively, while corresponding macro F1-scores were 82.89%, 81.39%, and 86.96%. We also observed that macro-averaged precision and recall improved as the proportion of training data increased.

To assess variability and statistical significance, 95% bootstrap confidence intervals were computed for each split. The intervals were relatively narrow, indicating that the Random Forest classifier produces stable predictions across different splits. Cohen's Kappa values were 0.860 for all splits, indicating strong agreement beyond chance. Permutation tests confirmed that the observed accuracies were highly significant ($p = 0.0005$ for all splits).

To further characterise the model's behaviour, we have reported per-class performance for the Random Forest classifier on the 80:20 test split. Regarding decision thresholds, the predicted label was assigned by applying the standard argmax rule to the output probability vector, without imposing any custom per-class thresholds.

Among the classes, the Non-demented class achieves the strongest per-class performance with a Precision of 0.960 and Recall of 0.974. The Demented class also performs strongly, with minor recall loss compared to the Non-demented class. The Converted class presents the classification records with lower precision (0.822) and recall (0.647) compared to the other two classes. The detailed results are reported in Table 8.

Further evaluation using stratified k-fold cross-validation on the 80% training set demonstrated that the model is robust with respect to the number of folds (Table 9). For k

= 3, 5, and 7 folds, the mean cross-validation accuracies were 0.888, 0.892, and 0.943, respectively, while the mean Cohen's Kappa values were 0.802, 0.807, and 0.922. Permutation tests conducted on the fixed 20% holdout set again confirmed statistical significance ($p = 0.0005$ for all k). These results indicate that increasing the number of folds generally improves the reliability of fold-wise performance estimates.

Analysis of misclassifications on the 80:20 test split showed that errors predominantly involved the Converted class. Cases misclassified as Non-demented exhibited higher MMSE (>28) and nWBV (>0.73), resembling healthy aging, while those misclassified as Demented showed lower MMSE (<26) and nWBV (<0.71). This bidirectional mismatch indicates that the Converted class occupies an intermediate feature space. Misclassifications between Demented and Non-demented groups were minimal ($<5\%$).

5.3 Decision tree rules interpretation

Decision trees offer deterministic inference and explicit logical rules. In this study, two widely adopted tree induction strategies were considered: CART, which optimizes Gini impurity, and C4.5, which maximizes information gain using entropy. The CART model produces compact binary trees with stable decision boundaries, whereas C4.5 allows more informative splits through entropy-based optimization. The feature space is restricted to clinically meaningful variables, including MMSE, SES, EDUC, Age, and nWBV. Quantitative performance on the held-out test set indicates that the C4.5 model achieves higher accuracy and macro-averaged F1-score than CART. The comparative results are reported in Table 10.

Table 10 Performance comparison of decision tree models on the test set

Model	Metric	Result
CART (Gini)	Accuracy	0.8873
	Accuracy CI	[0.8169 to −0.9577]
	Macro precision	0.7770
	Macro recall	0.7623
	Macro F1	0.7680
	Macro F1 CI	[0.6263 to −0.8928]
	Macro F1	0.9014
C4.5 (Entropy)	Accuracy	0.9014
	Accuracy CI	[0.8310 to −0.9577]
	Macro precision	0.8089
	Macro recall	0.8059
	Macro F1	0.8040
	Macro F1 CI	[0.6481 to −0.9180]
	Macro F1	0.8040

The best accuracy and macro F1-scores are bolded

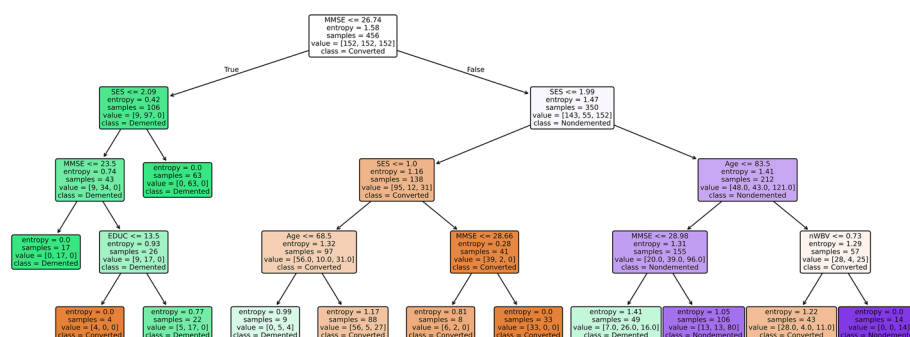


Fig. 5 Illustration of the C4.5 decision tree. The tree is visualized in 4 depths

We have further extracted the Decision tree rules for the learned C4.5 method. Algorithm 1 presents the decision tree rules, and Fig. 5 visualizes the decision tree (at 4 depth).

Input: patient features: MMSE, SES, EDUC, Age, nWBV
Output: predicted class $\in \{\text{Demented, Converted, Nondemented}\}$

```

1: if MMSE  $\leq 26.74$  then
2:   if SES  $\leq 2.09$  then
3:     if MMSE  $\leq 23.50$  then
4:       return Demented
5:     else
6:       if EDUC  $\leq 13.50$  then
7:         return Converted
8:       else
9:         return Demented
10:      end if
11:    end if
12:  else
13:    return Demented
14:  end if
15: else
16:   if SES  $\leq 1.99$  then
17:     if SES  $\leq 1.00$  then
18:       if Age  $\leq 68.50$  then
19:         return Demented
20:       else
21:         return Converted
22:       end if
23:     else
24:       return Converted
25:     end if
26:   else
27:     if Age  $\leq 83.50$  then
28:       if MMSE  $\leq 28.98$  then
29:         return Demented
30:       else
31:         return Nondemented
32:       end if
33:     else
34:       if nWBV  $\leq 0.73$  then
35:         return Converted
36:       else
37:         return Nondemented
38:       end if
39:     end if
40:   end if
41: end if

```

Algorithm 1 Decision Tree Rules

5.4 Model interpretability using SHAP

To further support model interpretability, we applied SHAP (SHapley Additive exPlanations) to the best-performing Random Forest model. SHAP is a model-agnostic interpretability method that quantifies the contribution of each feature to individual predictions. Using the trained Random Forest model and the held-out test set, SHAP values were computed to analyze both global feature importance and class-specific feature contributions.

Figure 6 presents the SHAP summary plots for the three diagnostic classes (Converted, Demented, and Nondemented). The results indicate that MMSE and normalized whole brain volume (nWBV) consistently exhibit the strongest influence on the model's predictions across classes, while Age, EDUC, and SES show moderate but consistent contributions.

These SHAP explanations provide an additional layer of interpretability by illustrating how individual features influence model predictions within the dataset.

5.5 Graph analysis

A graph is one of the most effective data structures to represent entity relationships and interactions [35]. Biological networks, transportation networks, and other systems can be represented as graphs. Each node represents an entity in a graph, while the edges represent the interactions between nodes. In this study, we used Neo4j (an open-source tool) to show connected patterns and present descriptive summaries from graphs in an understandable way. The tool uses a query language to extract and aggregate information. Importantly, all graph-based representations in this study are descriptive and exploratory in nature. They are intended to summarize observed group-level differences in the dataset and to complement the statistical and machine learning analyses. Observed associations between markers and diagnostic groups reflect statistical patterns within the OASIS-2 samples.

5.5.1 Clinical dementia rating and feature correlation

Mapping the number of patients according to the CDR value: Graph 1. Clinical Dementia Rating (CDR) is a significant marker for the condition of a dementia patient. Figure 7 presents the distribution of diagnostic group types across CDR values, based on real, non-synthetic samples. The three primary CDR nodes represent values 0, 0.5, and 1, each connected to its corresponding diagnostic group nodes.

For CDR = 0, patients are categorized as either Non-demented or Converted. The majority of individuals at this stage belong to the non-demented group, while a smaller subset is classified as converted, reflecting individuals who are clinically non-demented at baseline but may transition to dementia over time in longitudinal follow-up. For CDR = 1, all patients are classified as Demented, consistent with the clinical definition of this stage. No non-demented or converted cases are observed at CDR = 1, resulting in a single connection between the CDR = 1 node and the Demented group. Importantly, CDR = 0.5 is treated strictly as a mild cognitive impairment (MCI) or prodromal stage. As shown in Fig. 7, patients at this stage are predominantly classified as either Converted or Demented. This observation supports the interpretation of CDR = 0.5 as a transitional stage with a higher association with prodromal or early dementia profiles. In the subsequent figures, we further contextualize these groupings by examining how clinically relevant marker values vary across diagnostic groups within each CDR stage.

Exploration of group types correlation with CDR 0 and 1: Graph 2. As shown in Fig. 8, individuals with a CDR value of 0 may belong to either the Non-demented or Converted group. Using Cypher queries in Neo4j, we computed group-wise mean values of clinically relevant attributes derived exclusively from real samples. Each feature node is connected to its corresponding group node to facilitate a clear, interpretable comparison of feature distributions across groups.

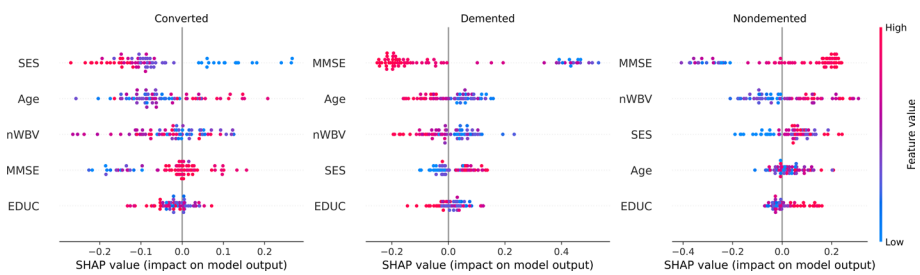


Fig. 6 SHAP summary plots showing the contribution of each feature to the Random Forest model predictions for the three diagnostic classes

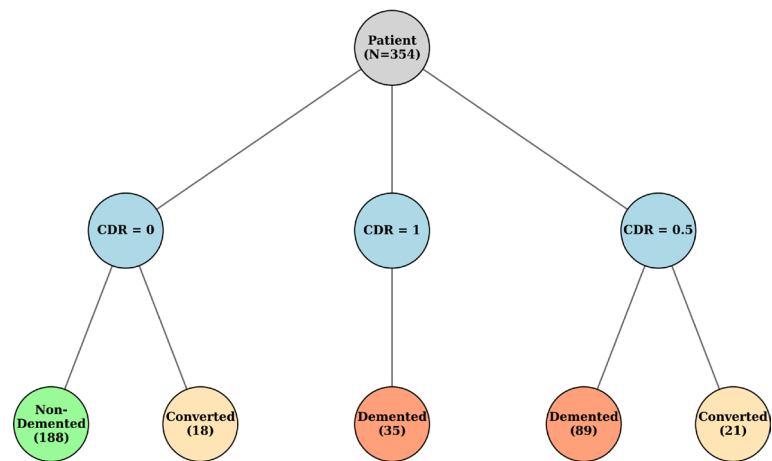


Fig. 7 Group type for corresponding CDR value

For the CDR 0 (Non-demented) group, the average values of MMSE, nWBV, EDUC, SES, and Age indicate comparatively preserved cognitive performance and brain volume. In contrast, the CDR 0 (Converted) group exhibits generally lower values for MMSE, nWBV, and SES, while age and education levels remain broadly comparable. These differences are consistent with a prodromal or transitional clinical profile and align with the trends observed in the probability density analyses. Importantly, this agreement with the PDF-based analysis serves as cross-method descriptive validation, not as evidence of causality or diagnostic hierarchy. All individuals with a CDR value of 1 are categorized as Demented, and their mean MMSE and nWBV values are lower relative to the other groups. The visualisation provides a group-level summary of marker distributions across CDR stages. Overall, the graph-based representation complements the statistical and machine learning analyses by offering a transparent, sample-level view of how marker values are distributed across diagnostic groups.

Impact on group types based on critical markers for the values of CDR 0.5: Graph 3. Figure 9 illustrates the distribution of clinically relevant feature averages for individuals classified under CDR 0.5.

It represents the mild cognitive impairment (MCI) stage and distinguishes between two subgroups: Converted (Prodromal) and Demented (Confirmed).

A patient with a CDR value of 0.5 represents the mild cognitive impairment stage and may clinically transition toward dementia. To examine group-wise characteristics at this stage, we used multiple Cypher queries to compute feature-wise average values for each subgroup based on real samples only.

The Converted (prodromal) group exhibits higher MMSE scores and slightly higher nWBV, suggesting relatively preserved cognitive function and brain structure. In contrast, the Demented group shows lower MMSE scores and reduced nWBV. Differences in Age, EDUC, and SES are observable but are interpreted as contextual characteristics rather than diagnostic drivers. These group-level distributions are consistent with the patterns observed in the probability density analyses, which similarly show intermediate marker values for the Converted group and lower mean MMSE and nWBV scores for the Demented group in this sample. Overall, the results are consistent with the characterization of CDR = 0.5 as a transitional stage, where marker distributions in this data set tend to sit between the non-demented and confirmed-demented profiles.

Table 11 presents a comprehensive summary of the mean $\pm$ SD values of all clinical markers across diagnostic groups and CDR stages. The table illustrates how marker values vary between group types as the CDR score progresses, enabling a direct comparison of feature trends across disease stages. Examining these differences provides insight into how specific markers are distributed across clinically meaningful groups.

Correlation analysis of selected features and highlighting the impact on group types: Graph 4. Figure 10 depicts the correlation between the variables. Our analysis uses Pearson's correlation.

We assess the linear relationships between clinically relevant variables using Pearson correlation, where values close to -1 indicate a strong negative linear association, values close to 0 indicate no linear association, and values close to 1 indicate a strong positive linear association. Only statistically significant associations with higher absolute correlation values are retained and represented as edges in the graph to improve interpretability.

The analysis reveals a positive association between MMSE and nWBV in this sample. Participants with a higher normalized brain volume tended to score higher on the MMSE. Age shows a moderate negative association with nWBV, with older participants tending to have lower normalized brain volume scores. A positive association is observed between EDUC and MMSE. SES and EDUC show a strong positive association with each other, consistent with their shared socioeconomic characteristics. A weaker association is present between SES and MMSE. Table 12 reports the topological centrality measures



Fig. 8 Estimated Features for CDR 0 and 1

of the clinically relevant feature association network constructed from real, leakage-free data. The centrality metrics quantify the structural roles of features within the undirected correlation network, rather than diagnostic or causal importance.

MMSE exhibits the highest betweenness centrality, indicating that it serves as a key connector between multiple feature groups in the network. This suggests that MMSE lies at the intersection of demographic and neuroanatomical variables, reflecting its role as a summary measure of cognitive status that covaries with several other features. On the other hand, EDUC and SES exhibit the highest eigenvector centrality values, reflecting their strong mutual association and connections to other influential nodes. This highlights the close socioeconomic linkage between education and socioeconomic status, consistent with their shared contextual influence rather than direct neurobiological effects. Additionally, nWBV demonstrates moderate betweenness and eigenvector centrality, positioning it as an important bridge between age-related and cognitive features. Lastly, Age displays comparatively lower centrality values, indicating a more peripheral role in the network. This supports its interpretation as a background risk factor that indirectly influences other features, rather than acting as a central associative hub.

5.5.2 Effect size analysis

To complement the descriptive statistics and graph-based visualizations, we quantified the magnitude of differences between diagnostic groups using Cohen's d effect sizes (Table 13). Unlike p -values, which are influenced by sample size, effect sizes provide a standardized measure of practical significance. Following Cohen's conventions as Petousis et al. [36], $d = 0.2$ is small, $d = 0.5$ medium, and $d = 0.8$ large.

The largest effects are observed between Demented and Non-demented groups for MMSE ($d = 2.60$) and nWBV ($d = 1.31$), confirming that cognitive performance and brain volume are the most powerful discriminators. For the Converted versus Non-demented comparison (prodromal stage), SES ($d = 0.80$) and MMSE ($d = 0.76$) show the largest effects, highlighting the role of socioeconomic factors and early cognitive changes. The Demented versus Converted comparison reveals large effects across multiple markers: MMSE ($d = 1.83$), Age ($d = 0.90$), SES ($d = 0.89$), and EDUC ($d = 0.75$), reflecting the multidimensional nature of dementia progression. These effect sizes align

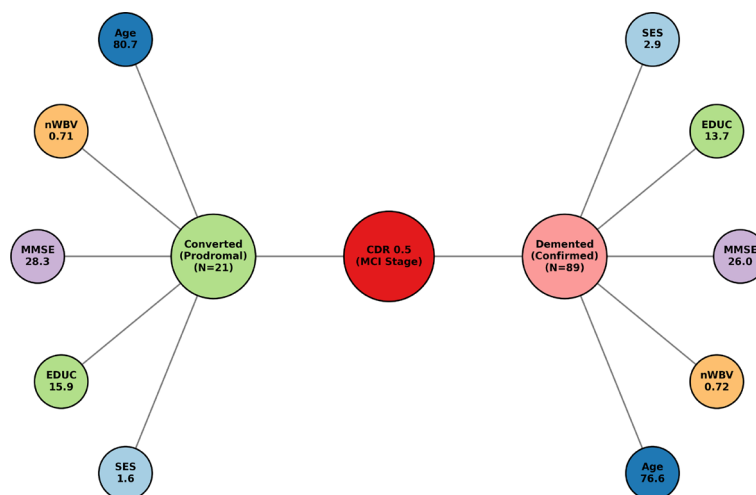


Fig. 9 Estimated Features for CDR 0.5

Table 11 Descriptive Statistics of Clinical Markers by CDR and Diagnostic Group (Mean ± SD)

Marker Name	CDR 0		CDR 1	CDR 0.5	
	Non-demented	Converted	Demented	Converted	Demented
EDUC	15.2 ± 2.7	15.0 ± 2.4	14.0 ± 2.5	15.9 ± 2.6	13.7 ± 3.2
nWBV	0.741 ± 0.038	0.735 ± 0.032	0.699 ± 0.022	0.713 ± 0.035	0.722 ± 0.034
MMSE	29.2 ± 0.9	29.1 ± 1.3	20.3 ± 5.3	28.3 ± 1.7	26.0 ± 3.1
SES	2.4 ± 1.0	1.8 ± 1.0	2.6 ± 1.2	1.6 ± 1.0	2.9 ± 1.2
Age	77.0 ± 8.1	78.7 ± 7.8	74.4 ± 6.9	80.7 ± 7.1	76.6 ± 7.2

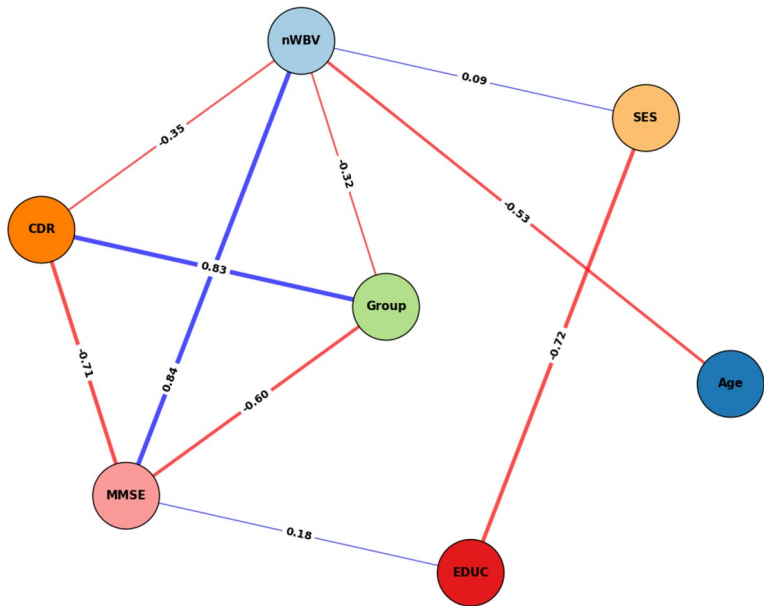


Fig. 10 Illustration of feature correlation

with the feature importance rankings from our machine learning models and the centrality metrics from our graph analysis.

6 Discussion

In this study, our objective was to develop an enhanced system for the early detection of dementia by integrating machine learning techniques with graph analysis, utilizing clinically significant markers. Our key findings unveil promising results in the realm of dementia diagnosis. While many prior studies using the OASIS-2 dataset have focused on improving predictive model performance, our study takes a different approach by integrating feature ranking with graph-based analysis. The use of graph visualization techniques, such as Probability Density Function (PDF) graphs and correlation matrices, provides a deeper understanding of the relationships between clinical markers and dementia stages. These visual tools not only enhance the interpretability of the results but also uncover hidden patterns in the data that may not be easily detected through traditional machine learning methods alone.

In this study, we evaluated ten machine learning models for dementia classification using clinical markers from the OASIS-2 dataset. Among these, the RF classifier achieved the highest performance with a test accuracy of 94.37% under an 80:20 split and SMOTE augmentation. The model has also shown strong statistical significance of a permutation *p*-value of 0.0005 and a Cohen’s Kappa of 0.922. Gradient Boosting and

Table 12 Topological centrality metrics of the clinically relevant feature association network

Node	Degree centrality	Betweenness centrality	Eigenvector centrality
EDUC	0.50	0.000	0.638
SES	0.50	0.000	0.631
MMSE	0.75	0.833	0.367
nWBV	0.50	0.500	0.207
Age	0.25	0.000	0.132

Centrality measures describe the structural roles of features within an undirected Pearson correlation network constructed from real, leakage-free data

Table 13 Effect sizes (Cohen’s *d*) for pairwise comparisons between diagnostic groups: Demented (Dem), Non-demented (ND), and Converted (Conv)

Feature	Dem versus ND	Conv versus ND	Dem versus Conv
MMSE	2.60	0.76	1.83
nWBV	1.31	0.76	0.50
Age	0.34	0.47	0.90
EDUC	0.46	0.26	0.75
SES	0.18	0.80	0.89

XGBoost models also performed competitively, with accuracies above 90% under optimal configurations. Through multi-strategy feature analysis, including Random Forest importance, Chi-square statistics, and Boruta ranking, we consistently identified MMSE and normalized whole brain volume (nWBV) as the most influential markers, while education (EDUC) and socioeconomic status (SES) provided contextual risk stratification. We have further explored decision tree methods such as the CART and C4.5 decision tree to enhance interpretability.

Our graph-based analysis provides a descriptive, sample-level visualisation of marker distributions and co-occurrence patterns across diagnostic groups. Graphs depicting CDR relationships showed that in this dataset, CDR = 0.5 was associated with both converted and demented groups, consistent with its clinical characterisation as a transitional stage. Correlation graphs showed a positive association between MMSE and nWBV and a strong co-variation between EDUC and SES in this sample.

Despite the complexity of the disease and the numerous contributing factors, our system demonstrated high accuracy, suggesting that the selected markers are associated with dementia classification within this dataset. This emphasizes the importance of the use of clinical expertise and AI-driven analysis in diagnostic processes. The implications of our findings are significant in the context of dementia diagnosis and treatment. By understanding the relationships between critical markers, our system facilitates early detection and provides insights into the disease. Furthermore, our approach emphasizes the importance of interdisciplinary collaboration between clinicians and data scientists. Integrating clinical expertise with AI-driven analysis enhances the accuracy and reliability of diagnostic models.

While the proposed framework demonstrates strong performance, certain limitations such as dataset dependency and class imbalance may influence generalizability. These aspects highlight opportunities for future work, including the integration of larger and more diverse datasets and the incorporation of multimodal data sources.

Furthermore, integrating multimodal data sources, such as neuroimaging and genetic markers, could provide a more comprehensive understanding of dementia pathophysiology. To efficiently process such expanding multimodal datasets, future work can utilize

scalable computational methods, including parallel in-memory processing and deep learning techniques [37, 38]. Advanced graph-based techniques, including dynamic graph convolutional networks, may also offer further insights into disease progression dynamics. Moreover, prospective longitudinal studies are needed to validate the efficacy of our proposed model in real-world clinical settings.

7 Conclusion

Dementia is a progressive and incapacitating neurological disorder that results in a gradual decline in cognitive function, including memory loss, impaired judgment, and language and communication difficulties. This disorder affects millions of people around the globe, and its prevalence is expected to increase in the coming years. Early diagnosis and treatment can help slow the disease's progression and enhance the quality of life for those affected.

In this study, we propose an integrative framework for dementia prediction that combines clinically validated markers with machine learning and graph-based analytical techniques. We evaluated multiple machine learning models to ensure robustness and reliability. We have further provided graph-based analysis to capture relational patterns among markers. To learn about the decision rules, we have included the CART and C4.5 decision tree methods. We have achieved the highest accuracy and F1-score of 94.37% and 86.96%, respectively. The model's confidence interval and statistical results remain consistent across different folds and split configurations. Extensive evaluation shows that our research may help improve the diagnosis of this serious condition.

Despite the promising results, our study has some limitations. The study is exploratory and methodological in nature, and the findings should be treated as such. The machine learning models presented here, while demonstrating high accuracy, were trained on a relatively small public dataset (OASIS-2), which limits the generalizability of the results. Furthermore, no external validation has been conducted with other datasets, which is a critical step in confirming the diagnostic accuracy and robustness of the models. It potentially limits the generalizability of our findings across diverse populations and dementia subtypes. Although we have used the SMOTE technique to balance the dataset and address this limitation, external and prospective validation using larger, more diverse datasets is required before any clinical interpretations or applications can be made. Until such validation is performed, these findings should be viewed as exploratory, offering insights into the relationships between key clinical markers rather than representing a validated diagnostic tool.

Author contributions

N.M.F. and S.S. contributed to the methodology. Conceptualization was carried out by S.A. and N.M.F. Formal analysis and investigation were conducted by N.M.F., D.S., M.A.R., and S.S. The original draft was prepared by N.M.F., D.S., M.A.R., and S.S., while M.J. and M.R.I. were responsible for review and editing. The research was supervised by S.A.

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Data availability

In this research, publicly available OASIS-2: Longitudinal MRI Data in Nondemented and Demented Older Adults dataset has been utilized. The dataset is accessible via this link: <https://sites.wustl.edu/oasisbrains/home/oasis-2>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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